

Hypervisor-Based Secure Virtualization Environment Deployment

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1. Project Overview

During my cybersecurity internship, I participated in the design and deployment of a secure virtualization infrastructure.

The objective was to:

- Build a reliable on-premises virtualization server
- Ensure storage redundancy and data protection
- Deploy multiple security and business VMs
- Integrate threat intelligence into the SIEM environment
- Prepare the environment for production and customer workloads

This project was executed in a **real-time enterprise environment** under the guidance of senior engineers.

2. Architecture Summary

Physical Storage Configuration

Component	Configuration
Base OS Storage	3 TB + 1 TB (RAID 1 → 1 TB usable)
Virtualization Storage	4 TB + 4 TB (RAID 1 → 4 TB usable)
Filesystem	ZFS via Cockpit
Purpose	High availability and data integrity

3. Storage Design

3.1 Base OS RAID

- **Physical Disks:** 3 TB + 1 TB
- **RAM:** 32 GB
- **RAID Level:** RAID 1 (Mirroring)
- **Usable Space:** ~1 TB

- **Purpose:**
 - Hypervisor base OS
 - System stability
 - Boot reliability

Ensures OS survives single disk failure.

3.2 Virtualization Storage Pool

- **Physical Disks:** 4 TB + 4 TB
- **RAID Level:** RAID 1
- **Filesystem:** ZFS
- **Usable Capacity:** ~4 TB
- **Managed via:** Cockpit Web Console

Purpose:

- VM storage
- Customer workloads
- Production services
- Snapshot capability

ZFS provides:

- Data integrity checks
 - Snapshots
 - Compression
 - Self-healing capability
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4. Hypervisor Environment

The server was configured as a **virtualization host** using Linux-based hypervisor management.

Key Responsibilities

- Hypervisor deployment
 - Storage pool creation
 - Network configuration
 - VM provisioning
 - Resource allocation
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5. Virtual Machine Deployment

The following virtual machines were created and configured.

VM1 – HRMS Server

Purpose: Human Resource Management System (Production)

Configuration:

- vCPU: 16
- RAM: 16 GB
- Storage: 1 TB
- OS: Ubuntu

Role:

- Internal HR operations
 - Employee data management
 - Business application hosting
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VM2 – Wazuh SIEM

Purpose: Security monitoring and threat detection

Configuration:

- vCPU: 16
- RAM: 8 GB
- Storage: 2 TB
- Role: Central SIEM

Functions:

- Log collection
 - Threat detection
 - Security alerting
 - Compliance monitoring
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VM3 – n8n Automation Server

Purpose: Security and workflow automation

Configuration:

- vCPU: 4
- RAM: 4 GB
- Storage: 500 GB

Use Cases:

- Alert automation
 - Workflow orchestration
 - Integration pipelines
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VM4 – MISP Threat Intelligence

Purpose: Threat intelligence platform

Configuration:

- vCPU: 4
- RAM: 8 GB
- Storage: 200 GB

Functions:

- IOC management
 - Threat sharing
 - Intelligence enrichment
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6. Wazuh–MISP Integration

One of the key achievements of this project was integrating **Wazuh SIEM** with **MISP Threat Intelligence**.

Objective

To enhance SIEM detection capability by enriching alerts with threat intelligence indicators.

Integration Workflow

1. Security events generated in environment
 2. Wazuh collects and analyzes logs
 3. Custom integration queries MISP
 4. IOC matching performed
 5. Enriched alerts generated in Wazuh
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Benefits Achieved

- Improved threat visibility
- Faster incident triage
- IOC-based detection
- Automated intelligence enrichment

- Better SOC efficiency
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7. Network and Management

Management Tools Used

- Cockpit (web-based server management)
- Linux CLI
- Hypervisor console

Key Activities

- VM networking configuration
 - Resource allocation tuning
 - Storage mounting
 - Service validation
 - Health monitoring
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8. Security Considerations

The environment was designed with the following security best practices:

- RAID 1 redundancy
 - ZFS data integrity
 - Segregated VMs
 - SIEM monitoring
 - Threat intelligence integration
 - Controlled administrative access
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9. My Role and Contributions

As a **Cyber Security Intern**, I worked with my team and contributed to:

- Assisting in RAID configuration
- Hypervisor setup support
- ZFS storage pool creation via Cockpit
- VM provisioning and resource allocation
- Ubuntu server installation
- Wazuh deployment support
- MISP deployment support
- Wazuh–MISP integration assistance
- Basic validation and health checks
- Documentation preparation

Note: Activities were performed under supervision in a production-like environment.

10. Key Skills Demonstrated

- Virtualization fundamentals
 - Linux server administration
 - RAID and storage planning
 - ZFS filesystem management
 - SIEM deployment (Wazuh)
 - Threat intelligence (MISP)
 - Security automation concepts
 - Infrastructure documentation
 - Team collaboration
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11. Conclusion

This project provided hands-on experience in building an enterprise-grade secure virtualization environment. The deployment successfully combined infrastructure reliability, security monitoring, and threat intelligence integration.

The experience strengthened my practical understanding of:

- SOC infrastructure
 - SIEM architecture
 - Production server deployment
 - Storage resilience
 - Security ecosystem integration
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